Technical Report: Periodic-1D Rank-One Single-Band Fourier Locality

Project Koios

Provisional non-retained development calculation, 29 October 2026

1 Scientific question and evidence boundary

For the lowest ordered eigenvalue field of the exact controlled four-state periodic parent, how do finite interpolation, deliberate Fourier-support truncation, and direct Hermitian scalar fitting behave as the construction mesh and retained hopping range are increased?

This report records a local, non-retained development calculation using the frozen finite controls. The calculation result was reconstructed by the independent verifier, all 725 named verification comparisons passed, and the authenticated two-figure renderer completed in memory. No execution directory, sealed evidence package, release, publication artifact, continuum result, uncertainty estimate, or material validation was created. The numerical statements below are therefore provisional finite-representation results rather than promoted project evidence.

2 Rank-one structural boundary

Let $H(q) \in \mathbb{C}$ be the scalar represented Hamiltonian and let $U(q) \in U(1)$. Scalar multiplication commutes, so

$$U(q)^\dagger H(q) U(q) = H(q) U(q)^\dagger U(q) = H(q). \tag{1}$$

An internal scalar phase therefore cannot redistribute the represented Hamiltonian among Fourier hopping coefficients. This calculation tests scalar interpolation, truncation, and fit behavior. It does not test the gauge-dependent represented-operator locality possible at rank two and above.

3 Exact parent and finite controls

The typed parent identity is `ksdft2effmass.periodic1d.controlled-four-state-parent.v1`. It is bound in the configuration wire to

$$H(q) = \lambda_0(K(q)), \qquad K(q) = T_0 + e^{2\pi i q} T_{+1} + e^{-2\pi i q} T_{+1}^\dagger, \tag{2}$$

with

$$T_0 = \begin{pmatrix} -6/5 & 3/50 & 0 & 0 \\ 3/50 & -7/20 & 0 & 0 \\ 0 & 0 & 1 & 1/25 \\ 0 & 0 & 1/25 & 7/4 \end{pmatrix}, \qquad T_{+1} = \begin{pmatrix} 3/20 & 0 & 3/25 & 0 \\ 0 & -1/10 & 0 & 1/10 \\ 0 & 0 & 1/20 & 0 \\ 0 & 0 & 0 & -1/25 \end{pmatrix}. \tag{3}$$

The entries are exact rational multiples of $E_{\mathrm{ref}} = 1$. Construction sizes are $N \in \{32, 64, 128, 256\}$, with $N = 256$ designated production. The evaluation inventory contains 1025 points

$$q_i = -\frac{1}{2} + \frac{i + 1/257}{1025}, \qquad i = 0, \ldots, 1024, \tag{4}$$

constructed as exact rationals before conversion to binary64. Retained ordinary supports are $R = 0, \ldots, 15$. The sampled-gap threshold is $0.25E_{\mathrm{ref}}$, the application error threshold is $10^{-8}E_{\mathrm{ref}}$, and the final adjacent-mesh refinement tolerance is 10^{-10} on the declared channel vector.

For each construction mesh, the complete cyclic interpolant I_N, transform-mediated truncation $T_{N,R}$, and direct Hermitian scalar fit $D_{N,R}$ are compared on the disjoint evaluation inventory. The reported channels include

$$e_{\mathrm{int}}(N) = \max_q |H(q) - I_N(q)|, \tag{5}$$

$$e_{\mathrm{trunc}}(N, R) = \max_q |I_N(q) - T_{N,R}(q)|, \tag{6}$$

$$e_{\mathrm{med}}(N, R) = \max_q |H(q) - T_{N,R}(q)|, \tag{7}$$

$$e_{\mathrm{dir}}(N, R) = \max_q |H(q) - D_{N,R}(q)|. \tag{8}$$

The producer calculates the result once. Independent verification reconstructs the parent eigensystems, Fourier sums, direct least-squares systems, diagnostics, masks, threshold decisions, refinement inventory, sampled-gap minimizer, and visualization projection without invoking the production analyzer.

4 Finite numerical results

4.1 Sampled gap and qualification disposition

The minimum sampled first-to-second parent eigenvalue gap was

$$\Delta_{\min} = 0.37249453282422684\, E_{\mathrm{ref}}. \tag{9}$$

It occurred on the production construction mesh at index 128, $q = 0$. The value exceeds the declared sampled threshold $0.25E_{\mathrm{ref}}$. This is a qualification over the finite sampled inventories only; it is not a global isolation certificate.

The first range at which both transform-mediated and direct-fit total maximum errors were at or below $10^{-8}E_{\mathrm{ref}}$ was $R = 11$ for every construction mesh. Passage remained sustained through $R = 15$. The maximum final $N = 128$ to $N = 256$ channel change was $6.2211547825082476 \times 10^{-15}$, below the declared 10^{-10} tolerance. The resulting bounded finite qualification disposition was *pass*. That disposition does not imply continuum convergence or scientific validation.

4.2 Mesh-dependent interpolation and representation diagnostics

Table 1: Finite construction-mesh diagnostics in units of E_{ref}.

N	reconstruction max.	Hermiticity max.	interpolation max.	interpolation RMS	first pass
32	4.441×10^{-16}	1.310×10^{-18}	2.234057×10^{-11}	1.077042×10^{-11}	11
64	4.449×10^{-16}	9.136×10^{-18}	8.882×10^{-16}	2.903×10^{-16}	11
128	4.458×10^{-16}	1.676×10^{-17}	1.110×10^{-15}	3.052×10^{-16}	11
256	6.670×10^{-16}	2.815×10^{-17}	8.882×10^{-16}	3.121×10^{-16}	11

The interpolation maximum fell from 2.234057×10^{-11} at $N = 32$ to values near binary64 roundoff for $N \geq 64$. The adjacent-mesh maximum channel changes for $32 \to 64$, $64 \to 128$, and $128 \to 256$ were respectively $2.2339685656902475 \times 10^{-11}$, $4.17817502814292 \times 10^{-15}$, and $6.2211547825082476 \times 10^{-15}$. These are finite adjacent-mesh diagnostics, not a continuum error estimate.

Every attained production support was nondominated under the declared (R, e_{med}) ordering because the mediated total maximum error decreased strictly over $R = 0, \ldots, 15$. No production supports formed a near-tie under the mixed 10^{-12} comparison rule.

4.3 Production support and error table

Table 2: Production $N = 256$ range diagnostics in units of E_{ref}. Displayed values are rounded; typed result bytes retain binary64 values.

R	e_{trunc}	e_{med}	e_{dir}	omitted L^2	coefficient defect L^2
0	2.969490×10^{-1}	2.969490×10^{-1}	2.969490×10^{-1}	2.094025×10^{-1}	4.441×10^{-16}
1	1.529255×10^{-3}	1.529255×10^{-3}	1.529255×10^{-3}	7.868300×10^{-4}	4.455×10^{-16}
2	4.702662×10^{-4}	4.702662×10^{-4}	4.702662×10^{-4}	2.415628×10^{-4}	4.462×10^{-16}
3	1.454887×10^{-4}	1.454887×10^{-4}	1.454887×10^{-4}	7.488650×10^{-5}	4.618×10^{-16}
4	4.475369×10^{-5}	4.475369×10^{-5}	4.475369×10^{-5}	2.309687×10^{-5}	4.498×10^{-16}
5	1.366150×10^{-5}	1.366150×10^{-5}	1.366150×10^{-5}	7.070482×10^{-6}	4.752×10^{-16}
6	4.136042×10^{-6}	4.136042×10^{-6}	4.136042×10^{-6}	2.146950×10^{-6}	4.673×10^{-16}
7	1.241321×10^{-6}	1.241321×10^{-6}	1.241321×10^{-6}	6.463664×10^{-7}	4.778×10^{-16}
8	3.691080×10^{-7}	3.691080×10^{-7}	3.691080×10^{-7}	1.928393×10^{-7}	4.717×10^{-16}
9	1.086649×10^{-7}	1.086649×10^{-7}	1.086649×10^{-7}	5.697590×10^{-8}	4.599×10^{-16}
10	3.164491×10^{-8}	3.164491×10^{-8}	3.164491×10^{-8}	1.665750×10^{-8}	4.590×10^{-16}
11	9.105412×10^{-9}	9.105412×10^{-9}	9.105413×10^{-9}	4.813878×10^{-9}	4.783×10^{-16}
12	2.584796×10^{-9}	2.584796×10^{-9}	2.584796×10^{-9}	1.373266×10^{-9}	4.830×10^{-16}
13	7.224646×10^{-10}	7.224646×10^{-10}	7.224648×10^{-10}	3.860166×10^{-10}	4.757×10^{-16}
14	1.982817×10^{-10}	1.982817×10^{-10}	1.982825×10^{-10}	1.066565×10^{-10}	5.409×10^{-16}
15	5.322764×10^{-11}	5.322764×10^{-11}	5.322831×10^{-11}	2.886786×10^{-11}	5.372×10^{-16}

At $R = 10$, both total maximum errors remained above the application threshold. At $R = 11$, the mediated and direct values were respectively $9.105412313026306 \times 10^{-9}$ and $9.105412868137819 \times 10^{-9}$, producing the first joint passage. The direct and transform-mediated coefficient vectors agreed to approximately 4.4×10^{-16}–5.4×10^{-16} in L^2 over the production range inventory; the maximum sampled route defect was $1.5544 \times 10^{-15} E_{\mathrm{ref}}$. This agreement is specific to the finite scalar control and the implemented binary64 routes.

The complete production coefficient magnitudes were symmetric to numerical precision. The dominant values were $|t_0| = 1.2116915623409472 E_{\mathrm{ref}}$ and $|t_{\pm 1}| \approx 0.1480688936028328 E_{\mathrm{ref}}$. Magnitudes then decreased from $5.295039809188736 \times 10^{-4}$ at $|R| = 2$ to approximately 7.26×10^{-11} at $|R| = 15$. The largest production Parseval absolute residual across the retained ranges was $1.3877787807814457 \times 10^{-17} E_{\mathrm{ref}}^2$.

5 Authenticated provisional figures

Figures 1 and 2 render only the authenticated projection from the passing in-memory verification. They are retained here as provisional report illustrations, not as an execution package, sealed evidence, or publication figure set.

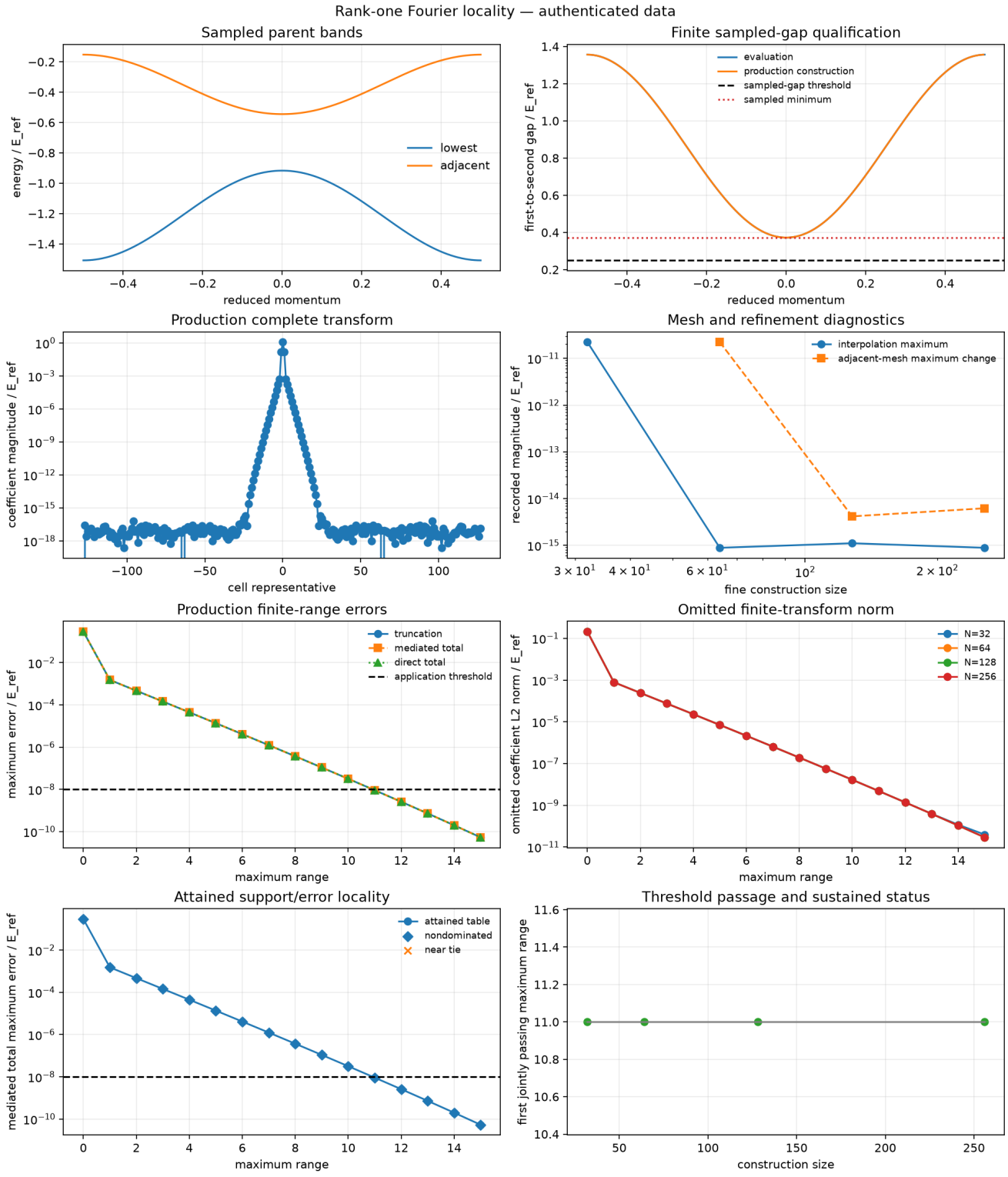


Figure 1: Authenticated finite rank-one Fourier-locality projection. The eight panels show the sampled parent bands; evaluation and production sampled gaps with their finite threshold; complete production coefficient decay; interpolation and adjacent-mesh refinement; production truncation, mediated-total, and direct-total errors; omitted coefficient norms; attained support/error masks; and first sustained threshold passage. All quantities are finite binary64 diagnostics.

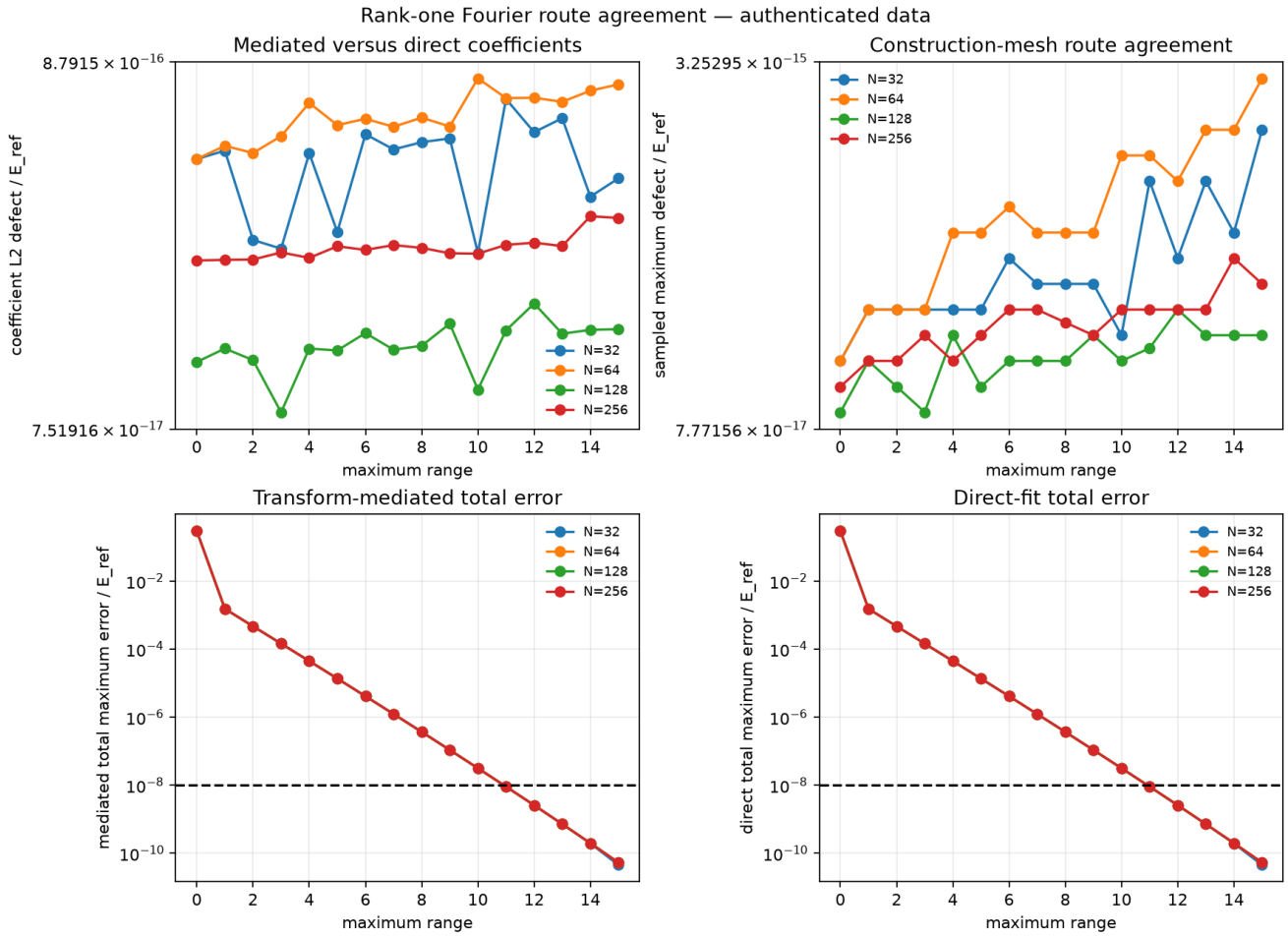


Figure 2: Authenticated finite route-agreement projection. The upper panels show the coefficient-vector and construction-sample defects between transform-mediated and direct Hermitian fits. The lower panels show the two total-error routes and the declared $10^{-8}E_{\mathrm{ref}}$ application threshold.

6 Independent verification and authenticated presentation

The byte-owning verifier reconstructed the typed result from the same immutable bytes used for its subject SHA-256. It independently regenerated the exact evaluation coordinates, bound parent identity, parent spectra, complete Fourier coefficients, truncated and direct candidates, error and norm diagnostics, design ranks and condition numbers, Pareto and near-tie masks, first and sustained passage, max/RMS triangle checks, refinement labels and values, sampled-gap minimizer and threshold side, finite qualification disposition, and every field of the visualization projection.

All 725 named observations passed. The largest observed discrepancy was $9.672262990534364 \times 10^{-12}$; its comparison remained within the applicable mixed tolerance, and the largest tolerance in the aggregate was $1.2691051231735601 \times 10^{-11}$. A passing software/numerical aggregate verifies the declared finite operations and correlations only. It does not establish a Formal certificate or scientific validation.

The authenticated renderer produced the two PNG values embedded in Figures 1 and 2. Their exact bytes are retained only as technical-report review illustrations. They were not copied into a retained execution, sealed evidence package, or publication asset inventory.

7 Provisional provenance

The development calculation used branch `work/periodic1d-rank1-control-v3` at base revision `bc1601f3a50dda19b994eea474ca522f2d836c26` in a dirty worktree. Runtime on the local development machine was approximately 1.12 seconds for calculation, 0.19 seconds for independent verification, and 0.88 seconds for rendering. Timings are descriptive, not benchmarks.

Exact SHA-256 identities from this non-retained development calculation were:

- configuration bytes: `5c9beb47eefe9749a249a74545a25c7ef8d3dd6eafec658d23dfbee574ace173`;
- result bytes: `3f24fbf6bd6d85040e98481b0487407932f43a70dd04690417396a50797fad2b`;
- verification bytes: `69232918f32f7006df0a5818925c35ff1462d4120448692aebdfc4a1489fd366`;
- visualization-data bytes: `5fd7fb9bf56c18cf112ae1060b672d1990c5cfc97661321eac4c735fdb816f3a`;
- locality report PNG: `df7032249383521b91012d18ef83025a72e6333b387e6b97b9acc898b2fdff62`;
- route-agreement report PNG: `c95f9ab1289baefffba637197d40218324f46c9b2b18476a8bccf2067352cf07`.

These hashes identify the exact development bytes only. They do not seal, promote, or publish them. No retained result or verification file currently exists in the repository.

8 Conclusions and limitations

Within the declared finite binary64 control, the lowest eigenvalue field has a rapidly decreasing scalar Fourier tail, construction meshes $N \geq 64$ agree on the evaluation inventory near roundoff, and both retained approximation routes first satisfy the $10^{-8}E_{\mathrm{ref}}$ application threshold at $R = 11$. The sampled gap and final adjacent-mesh stability gates pass, and all attained production supports are nondominated because error decreases strictly with range.

These conclusions are deliberately bounded. They do not establish a continuum limit, a global isolated band, uncertainty quantification, higher-rank gauge behavior, material applicability, publication acceptance, evidence sealing, or a Project Koios–Formal proposition. Retention or promotion requires a separately authorized direct execution and the repository's provenance and sealing gates.